

Large Language Model-based Web Navigating Agent

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Abstract

The integration of Large Language Models (LLMs) with web navigation capabilities represents a significant advancement in natural language processing. This field addresses the challenge of enabling AI systems to autonomously browse on the web, performing complex tasks like searching, summarizing, and data retrieval, similar to human users. This report focuses on autonomous web navigation agents, with particular attention to the recent developments in LLM-based approaches. Our goal is to examine the baseline performance of this model by reproducing its key results on the datasets and methods presented, and analyze its errors to suggest improvements for our final project.

1 Introduction

In recent years, Large Language Models (LLMs) have demonstrated impressive capabilities in natural language processing, extending their impact into tasks like web navigation, where users interact with online platforms through natural language instructions.

This project investigates the current state and future potential of LLM-based web navigation agents through three main components. First, we conduct a comprehensive analysis of the existing approaches in web navigation, focusing on the transition from traditional rule-based systems to modern LLM-powered agents. Second, we present our reproduction study of a competitive baseline model for web navigation. Through this reproduction, we aim to validate the reported performance metrics and identify specific areas where current approaches succeed or fail.

Third, building on our empirical findings, we analyze the limitations and errors of the current model, and propose several architectural improvements designed to address the identified limitations.

Through this work, we aim to contribute to the ongoing development of more robust and capable web navigation agents. Our findings suggest that while current LLM-based approaches show promise, significant opportunities exist for improving their reliability and effectiveness across diverse web environments.

2 Literature Survey

The field of autonomous web navigation has undergone significant evolution over the past decade, marked by several distinct phases of technological advancement. The earliest approaches, emerging around 2015 to 2018, focused primarily on supervised learning techniques. For instance, Web2Text(Vogels et al., 2018) navigates web elements by constructing neural network classifier using Convolutional Neural Networks (CNNs) to identify main content on web pages based on structural and textual features of consecutive text blocks. But they face significant challenges when dealing with dynamic layouts.

The field experienced a paradigm shift between 2019 and 2020 with the integration of reinforcement learning techniques. DOM-Q-NET(Jia et al., 2019) performs a novel architecture for reinforcement learning-based web navigation that uses graph neural networks to represent the tree-structured HTML of web pages, addressing the challenges of large discrete action spaces and varying numbers of actions between states.

The emergence of Large Language Models (LLMs) around 2020 marked another revolutionary change in web navigation systems. First-generation systems that incorporate LLM to web environments, such as WebGPT (Nakano et al., 2021), utilized pre-trained language models for page understanding and implemented question-answering(QA) tasks. Advanced LLM integration soon followed, combining language models

with visual understanding components and implementing few-shot learning for new tasks. For example, Mind2Web (Deng et al., 2024) emphasized complex tasks on websites, such as filtering webpage elements and selecting specific elements, while WebAgent (Gur et al., 2023) constructed web agent that learns from self-experience to complete user instructions on real websites by combining canonical web actions in a program space, integrating two LLMs for task planning, HTML summarization, and executable Python code generation. However, a primary concern is the computational overhead in task execution. While some systems, such as WebAgent (Gur et al., 2023) with its HTML-T5 and Flan-U-Plam architecture, have demonstrated impressive capabilities in web interaction, they face practical limitations due to their resource requirements. The use of large-scale models, particularly those in the 540B parameter range, creates substantial barriers to widespread deployment.

Therefore, our project aims to address challenges by focusing on developing a more efficient and deployable LLM-based web navigation agent. We propose to explore novel architectures that can maintain high performance while reducing computational requirements.

3 Baseline Model Reproduction

Our baseline implementation and experiment reproduction follows the experimental methodology outlined in AutoWebGLM (Lai et al., 2024).

3.1 Model Description

For our baseline model reproduction, we focus on three large language models: GPT-4o (OpenAI, 2024), LLaMA2-7B (Touvron et al., 2023), and ChatGLM3-6B (GLM et al., 2024). These models have been selected due to their competitive performance in a wide range of natural language processing tasks, and each chosen for specific reasons aligned with our experimental goals and computational constraints.

GPT-4o (OpenAI, 2024) was chosen as our primary baseline due to GPT-4’s (Achiam et al., 2023) superior performance demonstrated in the AutoWebGLM (Lai et al., 2024) study, where it achieved the highest success rates among the baseline models across various web navigation tasks. As a state-of-the-art language model, GPT-4o (OpenAI, 2024) serves as an improved upper

bound for performance benchmarking, providing insights into the maximum capabilities currently achievable in web navigation tasks.

Complementing this, we selected LLaMA2-7B (Touvron et al., 2023) as our second baseline model, representing a more lightweight alternative. With its significantly smaller parameter count (7 billion compared to GPT-4o’s estimated hundreds of billions), LLaMA2-7B (Touvron et al., 2023) represents an important point in the efficiency-performance trade-off spectrum. This choice was motivated by the growing need for computationally efficient solutions in practical applications, where resource constraints often play a crucial role in model selection.

Introduced as our third baseline model, ChatGLM3-6B (GLM et al., 2024) is the latest open-source model in the ChatGLM series. While retaining many excellent features such as smooth dialogue and low deployment threshold from the previous two generations, ChatGLM3-6B (GLM et al., 2024) introduces a powerful base model, employing a more diverse training dataset, more sufficient training steps, and a more reasonable training strategy. This model was selected to explore the performance of original baseline alternatives AutoWebGLM (Lai et al., 2024) which is based on ChatGLM-6B (GLM et al., 2024) in web navigation tasks, providing insights into the capabilities of models with a smaller parameter count and open-source availability.

3.2 Dataset Overview

Our task involves reproducing the evaluation metrics using these models on the MiniWoB++ (Humphreys et al., 2022) dataset, a commonly used benchmark for evaluating models on web navigation and interaction tasks.

MiniWoB++ (Humphreys et al., 2022) simulates real-world web interactions through carefully crafted scenarios such as clicking buttons, filling out forms, and extracting information from web pages. The dataset’s comprehensive coverage of web interactions from simple clicking tasks to complex multi-step operations aligns well with our research objectives, making it an ideal choice for assessing the capabilities of large language models to navigate web pages.

3.3 Framework Implementation

Our agents network depends on using LLM models as our backend processor and generate results

from reading the parsed html pages, here is the details of our implementation:

- **Instruction:** The agent is given a task or instruction, such as "Checkout my cart."
- **HTML (Parsed):** The state of the web page, provided to the agent as an HTML structure. The HTML may contain tags representing elements like navigation links, buttons, or forms, such as `<button[A]|home>` and `<input[A]|datepicker; text; hasDatepicker >`.
- **Previous Actions:** The list of prior actions the agent has taken. For example: `type_string("A", "8", False)` suggests the agent entered text 8 into an input field A without pressing return.
- **Prompt:** The given prompt combining the instructions and previously mentioned parts together is fed into the model for results.
- **LM Agent:** The LM (Language Model) agent uses the provided observation space, including instructions, HTML content, and previous actions, to understand the environment context and generate appropriate responses. The LM agent outputs the next action based on its decision-making. In a certain action space, it select a action like: `click("A")`.
- **Action Parser:** The given action is parsed by code and execute user-initiated interactions on the web page, such as clicking on elements, filling forms, or navigating links based on the context provided.

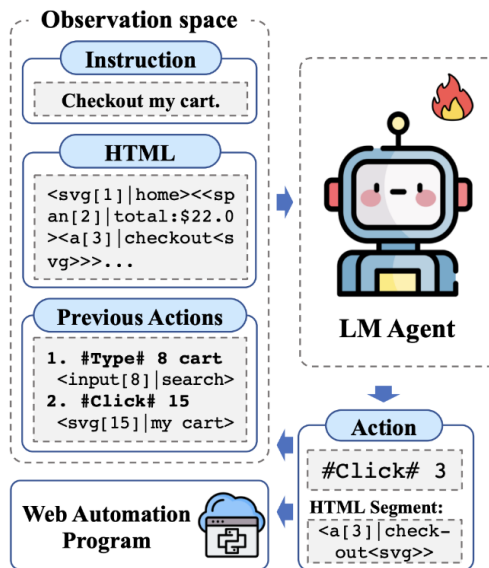


Figure 1: Agent Framework (GLM et al., 2024)

3.4 Experimental Setup

The evaluation of baseline models are performed on a T4 GPU within a Google Cloud Virtual Machine (VM). To reproduce the baseline results, we set up the environment as follows:

- **Model Selection:** We used the GPT-4o (OpenAI, 2024) model, the LLaMA2-7B (Touvron et al., 2023) and the ChatGLM3 (GLM et al., 2024) model for web navigation tasks.
- **Task Configuration:** Following the experimental setup from WebAgent (Gur et al., 2023), we test MiniWoB++ with 56 original tasks by running 10 evaluation episodes per task to evaluate model capabilities.
- **Evaluation Metrics:** The performance was evaluated based on the Average Score, which is the average of the rewards for each test case. Reward is a predefined value from the MiniWoB++ (Humphreys et al., 2022), it is initiated to 0 for each episode and calculated in the range (-1,1) during the process, presenting the values that the agent receives after taking an action in the state.

3.5 Results

We evaluated the performance of different models on the MiniWoB++ dataset, which comprises 56 interaction tasks designed to assess the web navigation capabilities of autonomous agents. Each task was executed using two loops to ensure consistency and reliability in the generated model outputs. Given the computational constraints, including model costs and runtime efficiency, we leveraged high-performance GPUs and closed-source models that is in our reach for generating these results. The primary focus was on assessing the baseline performance of several prominent Large Language Models (LLMs), particularly in terms of their web navigation proficiency. The results for each model, averaged across all tasks, are as follows:

Models	Accuracy
ChatGPT-4o	44.1%
Llama2-7B	23.21%
ChatGLM3-6B	14.3%

Table 1: Accuracy for MiniWoB++ tasks in Web Task Agent Performance

These results illustrate a marked disparity in performance across the different models. No-

tably, ChatGPT-4o outperformed the other models, achieving a score of 0.441, which highlights its comparatively robust web navigation and interaction capabilities. Llama2, with a score of 0.2321, displayed moderate competence but lagged significantly behind ChatGPT-4o, indicating potential areas for improvement, particularly in understanding and executing web-based tasks. ChatGLM3-6B, scoring 0.143, demonstrated limited efficacy in these tasks, suggesting that enhancements to its architecture or training approaches may be necessary.

The discrepancies in performance reflect the inherent challenges in enabling LLMs to autonomously navigate web environments. Factors such as context understanding, state tracking, and task-specific adaptations likely contribute to the observed variations. Future efforts will focus on analyzing task-specific errors to identify common failure modes, thereby guiding model improvements and the development of more capable web navigation agents.

4 Error Analysis of Baseline Model

Tasks can be categorized into clicking, text input, selection, and complex interaction. The error rates of each model in different tasks are shown in Table 2.

Each model demonstrates different strengths and weaknesses across tasks, with lower error rates reflecting better performance. ChatGPT-4o demonstrates better performance in handling more complex tasks, with the lowest error rate in Complex Interaction (44.44%) compared to Llama2-7B (88.89%) and ChatGLM-6B (100.00%). This suggests that ChatGPT-4o is more reliable in handling complex interactions, likely due to its ability to understand and execute multi-step actions accurately. It also shows moderate performance in the Select Task (56.25%), indicating a relatively good capacity for tasks that involve dropdowns. However, its relatively higher error rates in the Click Task (61.54%) and Input Task (66.67%) suggest that it may struggle with simpler tasks, where precision and direct interaction are essential, indicating space for improvement.

Llama2-7B performs moderately on complex interactions, achieving lower error rates in Select Task (81.25%) and Complex Interaction (88.89%) than ChatGLM but higher than GPT4-o. This indicates some ability in handling context-dependent

tasks that require multiple steps. Meanwhile, its error rate in Click Task is the lowest among the models (46.15%), suggesting its relative precision with simpler actions.

ChatGLM-6B demonstrates the worst overall performance, with consistently higher error rates across most tasks, especially in more complex tasks like Complex Interaction (100%) and Input Task (88.89%). This suggests low adaptability and without a comprehensive understanding of both straightforward and complex tasks, possibly due to its smaller size, a smaller proportion of web-pages in its training set, and its main capacity for bilingual conversation.

In summary, each model could improve with targeted training: ChatGPT-4o could enhance its precision in simpler tasks, Llama2-7B could refine its handling of multi-step complex actions, and ChatGLM-6B could benefit from improving both sides.

5 Proposed Improvements

To enhance the performance and adaptivity in web navigation tasks, future work could involve using language models specifically fine-tuned on HTML-based tasks. Also, we could fine-tune baseline models ourselves to create specialized models that better understand the structure and syntax of web elements. This approach is likely to improve the model's comprehension of HTML features and actions, leading to more accurate navigation.

Additionally, the current approach to converting language model output into actionable steps could be improved by reworking this translation framework. A modified framework that aligns LLM outputs more closely with explicit navigation actions can improve the possibility of ensuring each output directly corresponds to an action, thereby enhancing the precision of the agent's responses.

Meanwhile, to improve the model's interpretative accuracy, we propose revising the prompts fed into the language models, adopting a step-by-step prompt structure that emphasizes clear, sequential understanding of the web navigation task. Tailoring these prompts based on each model's specific strengths and weaknesses could lead to better outcomes over complex navigation requirements.

Lastly, beyond the current dataset, a broader evaluation across other benchmark datasets, such as AutoWebBench, MindToWeb, and others,

Task	ChatGPT-4o	Llama2-7B	ChatGLM-6B
Click Task	61.54%	46.15%	61.54%
Input Task	66.67%	77.78%	88.89%
Select Task	56.25%	81.25%	87.50%
Complex Interaction	44.44%	88.89%	100.00%

Table 2: Error Rates for Different Tasks across Models

would provide a more comprehensive assessment of the model’s web navigation capabilities. Testing across diverse datasets would help providing a more robust measure of the model’s effectiveness in handling a variety of web navigation tasks.

6 Conclusion

In conclusion, we focused on the performance on LLM-based web navigating models.

The models we selected and the final results we generated slightly differ from those in the original AutoWebGLM (Lai et al., 2024) study. The primary reason for this discrepancy is that we did not perform further fine-tuning on the LLaMA and ChatGLM models. This limitation was due to our restricted resources, which prevented us from fully completing the training and fine-tuning process. Additionally, we reached out to the authors of the original paper. But they informed us that the dataset, code, and results from their training process are not publicly available.

Despite this, we were still able to achieve significant results through our experiments and analysis. While the lack of fine-tuning and access to certain resources may have introduced some variations, the findings from our approach remain valuable and demonstrate the potential of the models we used.

Based on the analysis of web navigation performance across the three language models, GPT-4o (OpenAI, 2024), LLaMA2-7B (Touvron et al., 2023), and ChatGLM3-6B (GLM et al., 2024), we reach the observation that although GPT-4o demonstrated the most robust overall performance, this is largely due to its superior reasoning capabilities for complex tasks compared to the other baselines. However, it still requires improvement in terms of performance on simpler tasks. LLaMA2-7B performed moderately on multi-step complex tasks, showing room for improvement in handling more intricate processes. On the other hand, ChatGLM-6B showed the weakest performance among the three baselines, struggling both

with complex tasks and simpler ones. It needs significant enhancements in both areas to improve its overall effectiveness. These findings highlight the complexities of enabling LLMs to autonomously navigate the web, with factors such as context understanding and task-specific execution playing critical roles.

Regarding potential improvements, we have identified several areas for optimization within the web navigator’s structure. These include selecting a larger-scale training dataset, refining prompt generation, improving the comprehensibility of navigation steps, and enhancing the HTML-based parsing module. Through this study, we have gained a deeper understanding of the current state of LLM-based web navigating agents and the challenges they face, which lays the foundation for future research and development aimed at improving the effectiveness, adaptability, and overall performance of these models in complex web environments.

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